

Complete parameter estimates for models testing genetic and sex-specific effects on immune-predicted weight loss (n = 3

Model	Parameters	β	t	p	Model fit
Model 1: Genetic baseline $R^2 = 0.019$, $F(3,300) = 1.94$, $p = 0.122$, $AIC = 1419$		()			Residuals
	Intercept	9.570 (0.411)	23.27	<0.001***	
	HI ¹	-0.154 (0.666)	-0.23	0.818	
	He ²	-0.392 (2.035)	-0.19	0.847	
	HI × He	4.015 (3.436)	1.17	0.244	
		()			df: 300
Model 2: Main effects $R^2 = 0.063$, $F(4,299) = 5.05$, $p < 0.001$, $AIC = 1407$		()			Residuals
	Intercept	9.053 (0.372)	24.35	<0.001***	
	Sex (Male)	-0.090 (0.278)	-0.32	0.746	
	HI	0.185 (0.406)	0.46	0.649	
	He	1.556 (1.014)	1.53	0.126	
	Infection	1.125 (0.285)	3.94	<0.001***	
		()			df: 299
Model 3: Two-way interactions $R^2 = 0.093$, $F(10,293) = 2.99$, $p = 0.001$, $AIC = 1409$		()			Residuals
	Intercept	9.316 (0.514)	18.14	<0.001***	
	Sex (Male)	-1.308 (0.736)	-1.78	0.077.	
	HI	-0.670 (0.828)	-0.81	0.419	
	He	1.286 (2.379)	0.54	0.589	
	Infection	2.320 (0.795)	2.92	0.004**	
	Sex × HI	2.144 (0.833) ³	2.57	0.011*	
	Sex × He	0.547 (2.076)	0.26	0.793	
	Sex × Infection	-0.369 (0.571)	-0.65	0.519	
	HI × He	1.692 (3.508)	0.48	0.630	

Significance codes: ***p < 0.001, **p < 0.01, *p < 0.05, . p < 0.1
Models compared using likelihood ratio tests.

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Model	Parameters	β	t	p	Model Ii
	HI $\times$ Infection	-1.299 (0.841)	-1.55	0.123	
	He $\times$ Infection	-0.908 (2.095)	-0.43	0.665	
		()			df: 293

¹ HI = Hybrid Index (0 = pure *M. m. domesticus*, 1 = pure *M. m. musculus*)

² He = Expected heterozygosity = $2 \times \text{HI} \times (1 - \text{HI})$, measuring genetic admixture

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Significance codes: ***p < 0.001, **p < 0.01, *p < 0.05, . p < 0.1

Models compared using likelihood ratio tests.